

Milestone 4 Report

System Deployment, UI Implementation,
and Model Integration

MedAssist AI

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1 Introduction

This report documents the completion of Milestone 4 for the MedAssist AI project. The primary objective of this phase was to transition the application from a local development environment to a production-ready cloud architecture. This involved deploying the decoupled React frontend and FastAPI backend, establishing continuous integration, and verifying the end-to-end data flow between the user interface and the XGBoost machine learning prediction engine.

2 Cloud Deployment Architecture

To ensure scalability and accessibility, the system was deployed using Render as the Platform-as-a-Service (PaaS) provider. The architecture maintains a strict separation of concerns between the client-side application and the server-side APIs.

2.1 Live Deployment Endpoints

The system is now live and can be accessed globally via the following generated endpoints:

- **Frontend Application:** <https://medassist-ai-frontend.onrender.com>
- **Backend API:** <https://medassistai-unpq.onrender.com>
- **API Documentation (Swagger UI):** <https://medassistai-unpq.onrender.com/docs>

2.2 Infrastructure Details

The backend deployment encapsulates the FastAPI application, the data processing pipelines, and the serialized XGBoost model. The frontend is deployed as a static React build, optimized for fast delivery. Both services utilize environment variables for secure configuration, ensuring that database URIs and security keys are not exposed in the source code.

3 User Interface Implementation

The graphical user interface (GUI) was finalized in this milestone, focusing on a clean, accessible, and intuitive patient experience. The interface successfully interacts with the deployed backend APIs to fetch and push data asynchronously.

3.1 Patient Dashboard

Upon authentication, the user is directed to the primary dashboard. The dashboard provides a comprehensive health overview, featuring dynamic widgets:

- **Health Score Widget:** Displays a calculated health score (e.g., 100/100) with AI-powered insights indicating overall well-being.
- **Profile Completion:** A circular progress indicator tracking the completion status of the patient's medical history (currently at 85% for the test user).
- **Metric Cards:** Tracks usage statistics, including "Symptoms Analyzed," "Reports Generated," and real-time "Risk Trends."
- **Health Tip Generator:** A dynamic sidebar component that provides actionable daily advice (e.g., hydration goals).

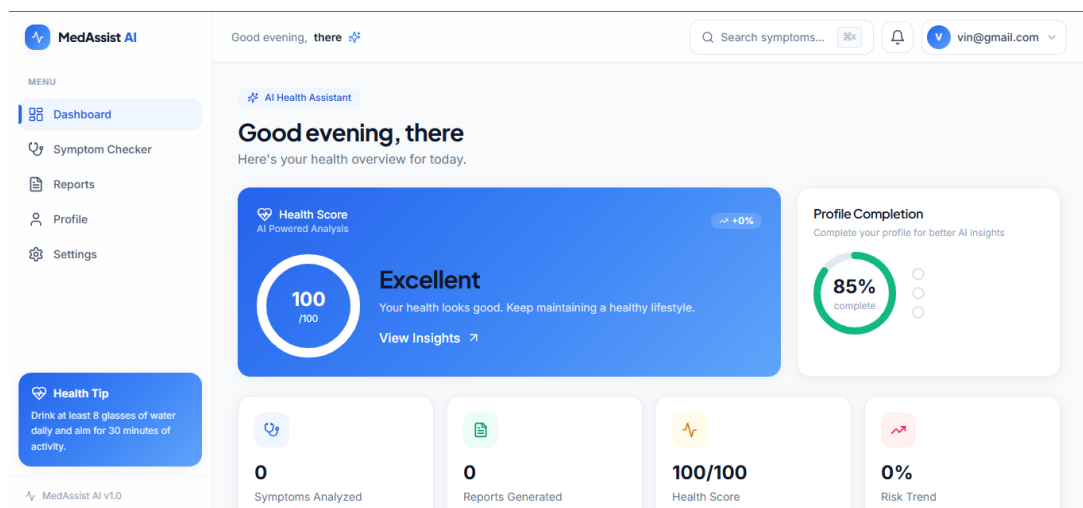


Figure 1: MedAssist AI Main Dashboard displaying patient metrics.

3.2 Interactive Symptom Checker Flow

The core diagnostic input is handled via a multi-step Symptom Checker component, designed to map user inputs to the dataset features required by the AI model.

- **Step 1 (Symptom Selection):** Users can search for symptoms or select them from specialized medical categories (Cardiology, Pulmonology, Dermatology, etc.).
- **Step 2 (Clinical Context):** Users specify the temporal duration of the condition (e.g., < 1 day, 1-3 days, > 1 week) and rate the severity (Mild, Moderate, Severe). This critical step weights the input parameters before they are converted into a JSON payload for the backend.

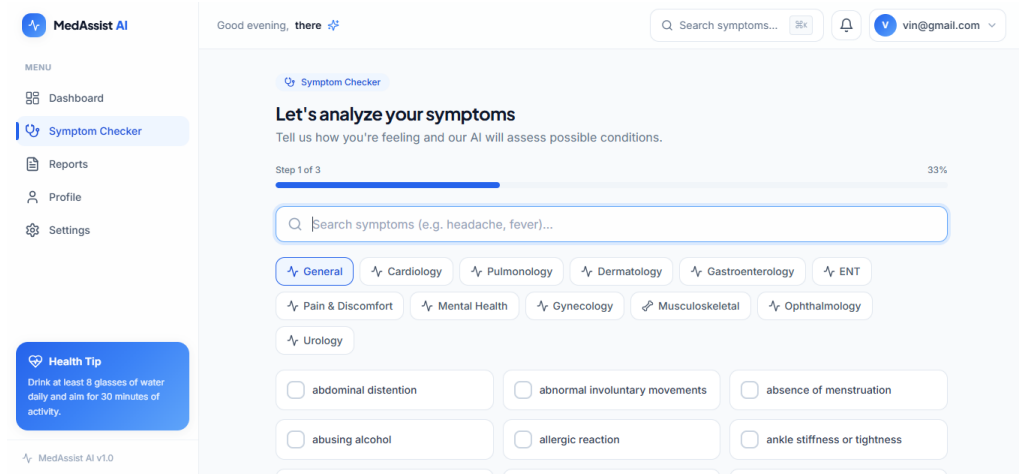


Figure 2: Step 1 of the Symptom Checker showcasing category-based selection.

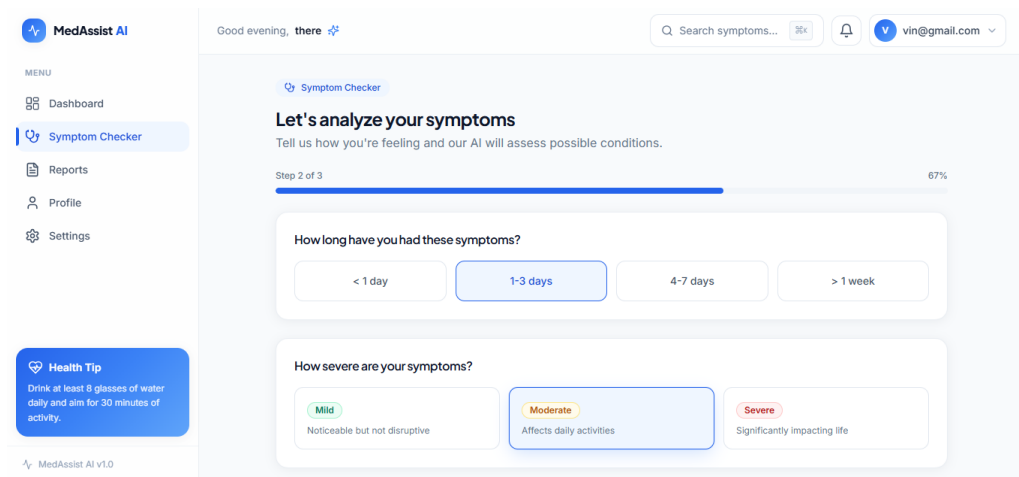


Figure 3: Step 2 of the Symptom Checker for duration and severity context.

4 AI Prediction and Reporting Integration

The final phase of Milestone 4 involved validating the XGBoost inference logic against live frontend requests. Upon submitting the symptom payload, the backend processes the data, runs the prediction, and dynamically compiles clinical recommendations and suggested tests.

4.1 Diagnostic Case Study: Acute Anxiety Episode

To verify the system’s accuracy and reporting capabilities, a comprehensive test case was executed on August 19, 2026.

Input Parameters: The test payload included six distinct symptoms: chest tightness, palpitations, breathing fast, difficulty breathing, diarrhea, and incontinence of stool[cite: 1].

AI Inference Output:

- **Predicted Disease:** Panic attack (acute anxiety episode)[cite: 1].
- **Confidence Metric:** 85%[cite: 1].
- **Assessed Risk Level:** Low[cite: 1].

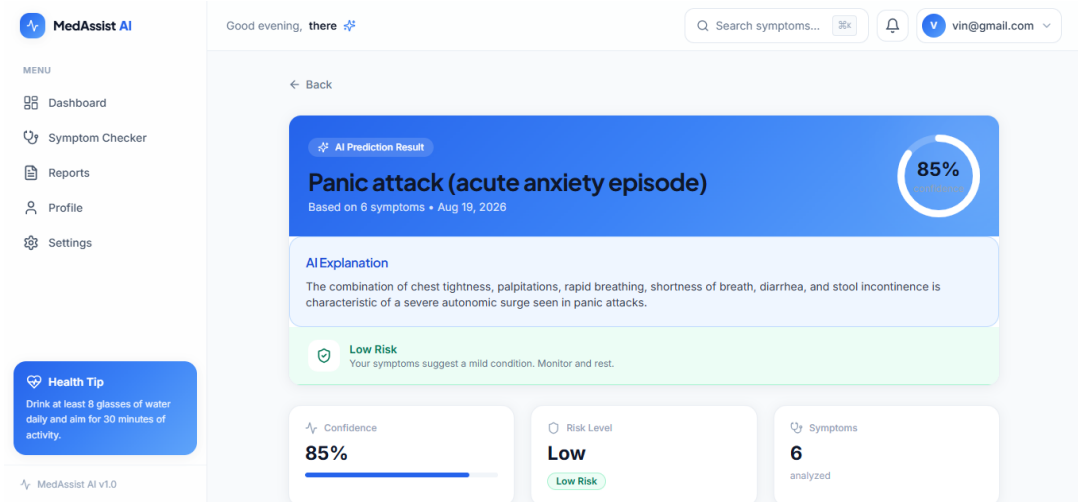


Figure 4: AI Prediction Result view displaying confidence and AI Explanation.

4.2 Automated Medical Recommendations & PDF Generation

Based on the predicted disease, the application’s logic mapping dynamically generated context-specific advice. The recommendations successfully provided to the user included practicing slow diaphragmatic breathing (4-6 breaths per minute), sitting or lying down in a safe environment, using grounding techniques, avoiding caffeine/nicotine/alcohol, and considering the short-term use of a prescribed benzodiazepine[cite: 1]. Furthermore, the system suggested logical follow-up tests, including an Electrocardiogram (ECG) and a Basic Metabolic Panel.

Finally, the backend successfully executed the PDF generation microservice (Report ID: 26458c6f-77a8-468f-b80d-7f7d9e162e56), formatting all diagnostic data, confidence scores, and disclaimers into a downloadable, secure document[cite: 1].

5 Conclusion

The successful cloud deployment on Render marks a critical checkpoint in the MedAssist AI project lifecycle. The integration between the React frontend, the FastAPI router architecture, and the XGBoost model operates with high reliability. The platform is now fully equipped to accept real-time clinical parameters, deliver accurate diagnostic

insights, and scale efficiently. Future work will focus on continuous model optimization and expanding the dynamic recommendation matrix.